

The mathematics of Markov models: what Markov chains can really predict in forest successions

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Abstract

The formalism of Markov chains is presented as a mathematical description of a succession process with a known scheme of successional transitions and their time characteristics. The basic hypotheses and assumptions behind the formalism are formulated and discussed from the viewpoint of the prognostic potential in the resulting models. Fundamental mathematical results from the theory of stochastic processes guarantee the existence of a stationary probability distribution of states in any finite, regular, time-homogeneous Markov chain, and any initial distribution of chain states does converge to some stationary distribution, matching the paradigm of classical succession theory. We propose a general method for estimation of time-homogeneous transition probabilities applicable for any kind of successional scheme, yet with strong requirements to the expert data: average duration times should be known for each specified stage of succession as well as the likelihood proportions among the transitions from the ramifying states of the scheme. Constructed by this method on a single set of data, the discrete- and continuous-time models are proved to converge to the identical distributions, while the average sojourn times in each state are shown to be also the same in the two models. Succession through forest types in a mixed (coniferous–deciduous) forest in Central Russia is considered as an example. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

When the course of succession in a plant community is described as a chain of discrete events each designating the onset of the subsequent stage in succession, the first step is made to build what

are nowadays called *Markov-chain models* of succession. These models have long been used to formalise the process of successional changes in terrestrial vegetation and to produce certain predictions in quantitative form. Publications on using this kind of models in ecological studies were booming in the 1970s and early 1980s (see reviews in Horn, 1975; Usher, 1979; Horn, 1981; Usher, 1981; Mirkin and Naumova, 1984), after which the boom was markedly eroded. Recent applica-

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tions however revived both practical interest in Markov models and theoretical debate on what these models can really predict in vegetation dynamics (see e.g. Usher, 1992; Acevedo et al., 1995; Li, 1995; Logofet, 1999; Balzter, 2000).

What a model can predict depends on what assumptions (or postulates) the model is based upon and in which way it is constructed. In the context of Markov-chain models this can be seen from the matrix of transition probabilities (or *transition matrix*), $P(s, t) = [p_{ij}(s, t)]$, the core of any chain model, whose entries show the probability that the chain, being in the state i at time s , will be in the state j at time t . When the state of the chain is registered at discrete instants of time, we deal with a *discrete* chain, and when the time variable is considered continuous, the chain is called *continuous*. The former may seem more natural to a field ecologist who observes the state of vegetation year by year, or in a longer time step. The latter however captures the natural course of time and has certain advantages for a modeller, which will be clear from the comparison of both approaches to be presented in this paper.

Relations between the discrete-time and continuous-time Markov processes is a theme that mathematical books could hardly leave untouched. However, traditional textbooks, like Doob (1953), Chung (1967), or Ventzel and Ovcharov (1991), treat the theme in a special context of discrete approximations for a given continuous process. This is different from the problem considered here: two kinds of processes constructed by the same method on a single data set.

We also analyse briefly the assumptions of *invariance* and *ergodicity*, whose acceptance or rejection underlies, either explicitly or implicitly, any case study applying a chain-modelling approach and determines the general properties of the resulting model including its prognostic ability. Furthermore, we present a rigorous analysis of the mathematical properties inherent in what are called *time-homogeneous* Markov chains, i.e. when the matrix $P = P(s, t)$, which is generally a function of both time instants t and s , actually depends only on the difference $t - s$. The outcome of the analysis suggests a mode of constructing

the transition matrix P from empirical evidence which does not require the ergodic hypothesis and which is quite different from what is proposed in the majority of landscape models relying on registration of the proper transitions in a set of observations (see e.g. Turner and Gardner, 1991; Vinogradov and Schitov, 1994).

We illustrate the analysis with an example of succession through forest types in a mixed (coniferous-deciduous) forest in Central Russia. Fig. 1 presents a scheme that describes, in the long term, the course of natural overgrowing after a tillage area has been abandoned — a scenario of forest secondary succession typical for the land-use history in the region (Korotkov, personal communication). Since no disturbances are considered in this scheme, the general direction of changes is unique: from the pioneer stage of succession to the climax, which is considered to be a polydominant spruce-broad-leaved forest (Popadyuk et al., 1994). Expert knowledge of how long a stage of succession can last (numbers of years in boxes of the scheme) is gained from both direct observations and available historical data.

2. General formalism of Markov-chain models for successions

To construct a Markov chain, one identifies the states of the chain with the specified stages of succession and enumerates them by natural numbers from 1 to K , as shown in Fig. 1. The set $S = \{1, \dots, K\}$ is thereafter the set of all the chain states, or stages of succession. Although the order of stages in this list may be arbitrary, marking the pioneer stages with the lowest and the final stages with the largest numbers has certain technical convenience. Let $X(t) \in S$ indicate the state at time instant t and thus serve, with the course of time t , a model representation for the course of succession. Process $X(t)$ is called a *Markov chain* when the following property holds true: if the chain is in a state i at a given time moment s , then the probability $p_{ij}(s, t)$ that it will be in a state j at a subsequent time instant $t > s$ does not depend on the chain behaviour before the moment s (the *Markov property*, see e.g. Karlin, 1968).

In formal terms, for any sequence of time moments $t_1 < t_2 < \dots < t_{n-1} < t_n$, we should have

$$\Pr\{X(t_n) = i_n | X(t_1) = i_1, \dots, X(t_{n-1}) = i_{n-1}\} \\ = \Pr\{X(t_n) = i_n | X(t_{n-1}) = i_{n-1}\}. \quad (2.1)$$

Thus, the conditional probability $\Pr\{X(t) = j | X(s) = i\} = p_{ij}(t, s)$ by definition. Arranged in a 2D-array, the transition probabilities $\{p_{ij}(s, t) | i, j = 1, \dots, K\}$ compose the *transition probability matrix*, of size $K \times K$, denoted normally by $P(s, t)$.

The question may arise why we are speaking of probabilities while the essence of classical succession theory is a regular (deterministic) nature of any successional dynamics. The logical answer would be because we normally see, when observing particular realisations of the ‘regular’ dynamics, quite a variety of succession trajectories from the pioneer stage to the climax one(s), and which particular trajectory is realised in a particular case may depend on a number of uncertain terms (like availability of proper seeds, variations in whether conditions, etc.). Even this alone makes room for stochastic description of the course of succession. Besides, whether or not a transition occurs in a fixed time interval is also a matter of probability,

even in the case where the scheme of transitions is non-ramified, unlike in Fig. 1 (see, e.g. Denisenko et al., 1996, for succession in the forest-steppe zone).

If the hypothesis of invariant environment conditions is adopted, then the Markov chain can be considered *time-homogeneous*, i.e.

$$p_{ij}(s, t) = p_{ij}(s - t), \quad \forall i, j \in S.$$

This means that each transition probability can only depend upon the length of the time interval in which it should act, rather than the actual time instant at which it is considered. Speaking in ecological terms, the *invariance* hypothesis should mean that no important changes occur in the key factors of the environment affecting the course of succession during the period of prediction as compared with the factors during the period of observations in which estimation of the transition probabilities can be relied upon.

Whether or not the invariance hypothesis can be adopted is a matter of formulation for each particular modelling case study. Here we concentrate on the mathematical benefits to be gained from the invariance hypothesis. It will be seen that all the mathematical properties of the chain

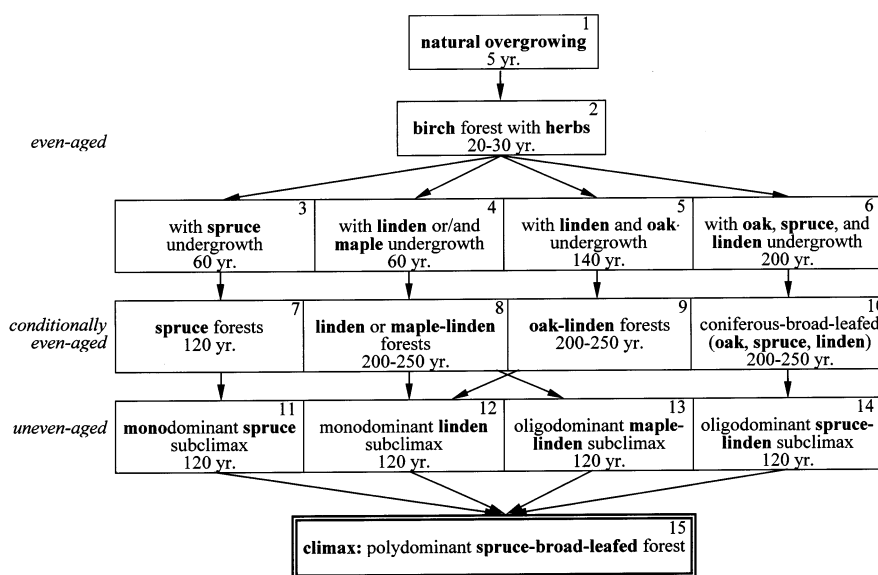


Fig. 1. Ideal course of succession through forest types (after-tillage series) in Prioksko-Terrasnyi Biosphere Reserve, Moscow Region, Russia (Korotkov, V.N., personal communication).

model, including its predictions, are entirely determined by the transition probability matrix. In Section 7 we discuss what might be done when the invariance hypothesis is inappropriate.

Another major assumption to be or not to be behind any quantified Markov-chain model is the so-called *ergodic hypothesis*, and we shall also discuss it in relation to the mode of estimating the transition probabilities from empirical evidence.

From the modelling viewpoint, the statistical properties of process $X(t)$ are of practical importance. In particular, the dynamics of vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_K(t)]$, which is the finite distribution of K probabilities $x_i(t)$, each meaning the probability, $P\{X(t) = i \mid \mathbf{x}(0)\}$, that the chain is in state i at time t , given an initial distribution $\mathbf{x}(0)$. The basic (matrix-vector) equation for $\mathbf{x}(t)$,

$$\mathbf{x}(t + \Delta t) = \mathbf{x}(t)P(\Delta t) \quad (2.2)$$

ensues from the probability meanings and the formal definitions adopted, while more intricate properties of $\mathbf{x}(t)$, in particular, the existence and stability of a limiting distribution $\mathbf{x}(\infty)$ or how long, on the average, it takes to reach certain states, were established in the theory of finite Markov chains, at least in the homogeneous case (see, e.g. Kemeny and Snell, 1960; Prokhorov and Rozanov, 1987). The limiting distribution(s) and the average times are usually considered to be the major outcome of the mathematical analysis following the estimation of transition probabilities.

Theoretically, further formalisation and analysis are possible along the following two approaches: (a) time is discrete and process $X(t)$ is observed only in certain time steps, i.e. $t = 0, 1, 2, \dots$, or (b) time is continuous ($0 \leq t < \infty$) and process $X(t)$ is observed at all t .

We proceed along the both approaches, assuming that (a1) the average waiting time, m_i , till the state i changes into another one is known for each state of the chain, and (a2) if a state can change directly into more than one others (ramification of the scheme), then all these transitions have known weight proportions among them (e.g. are equiprobable in a particular case).

The values of m_i could be gained from the expert knowledge of the time period that the forest remains in the corresponding state i , while

the knowledge of which transition is more likely might be formalised in the above ‘weight proportions’. Assumptions (a1) and (a2) are made to substantiate the mode of estimating the transition probabilities that will ensue from our analysis. In what follows, we express m_i in time steps.

3. Discrete model formalism

The transition probabilities require the following conditions to hold for any time instant $n = 0, 1, 2, \dots$:

$$p_{ij}(n) \geq 0, \quad i, j = 1, \dots, K \quad (3.1)$$

$$\sum_{j=1}^K p_{ij}(n) = 1, \quad i = 1, \dots, K \quad (3.2)$$

which are called the (row) *stochastic matrix* property. The Markov property (2.1) implies that, for any two time instants n, m , we have

$$P(n + m) = P(n)P(m) \quad (3.3)$$

The Kolmogorov–Chapman equation, with the initial condition:

$$P(0) = I \quad (3.4)$$

where $I = \text{diag}\{1, \dots, 1\}$ designates the diagonal (*identity*) matrix of the proper size.

In particular, $P(n + 1) = P(n)P(1)$ and Eq. (2.2) hence transforms into

$$\mathbf{x}(n + 1) = \mathbf{x}(n)P(1) \quad (3.5)$$

a system of first-order, linear difference equations with a constant matrix $P(1) = P$ of one-step transition probabilities.

The initial-value problem for Eqs. (3.3) and (3.4) can be easily shown to have the solution

$$P(n) = P^n \quad n = 0, 1, 2, \dots \quad (3.6)$$

and that for Eq. (3.5) to have respectively

$$\mathbf{x}(n) = \mathbf{x}(0)P^n \quad n = 0, 1, 2, \dots \quad (3.7)$$

Given an initial distribution of state probabilities $\mathbf{x}(0)$ — identified usually with the relative area distribution of the corresponding types in a mosaic of vegetation — Eq. (3.7) turns calculation of the model distribution at any given time instant into a matter of simple matrix multiplication.

Now we consider how the assumption (a1) enables one to use the available data to calculate the probabilities p_{ij} of transitions from state i into j in one time step. Let random variable W_i be the *waiting time* of $X(t)$ in a state i ($i = 1, \dots, K$), i.e. the time the chain has been waiting until the state changes into another one. Let the law of probability distribution for W_i be given by function $F_i(n) = \Pr\{W_i \leq n\}$, $n = 1, 2, \dots$. According to the definition, we have

$$\begin{aligned} 1 - F_i(n) &= \Pr\{W_i > n | X(0) = i\} \\ &= \Pr\{X(1) = X(2) = \dots = X(n) = i | X(0) \\ &= i\} \end{aligned}$$

which, by the Bayes formular of conditional probabilities and Markov property (Eq. (2.1)), obeys the following chain of equalities.

$$\begin{aligned} \Pr\{X(1) = X(2) = \dots = X(n) = i | X(0) = i\} \\ &= \frac{1}{\Pr\{X(0) = i\}} \Pr\{X(0) = X(1) = \dots \\ &= X(n) = i\} \\ &= \frac{1}{\Pr\{X(0) = i\}} \Pr\{X(n) = i | X(0) = \dots \\ &= X(n-1) = i\} \cdot \Pr\{X(0) = \dots \\ &= X(n-1) = i\} \\ &= \frac{p_{ii}}{\Pr\{X(0) = i\}} \Pr\{X(0) = \dots \\ &= X(n-1) = i\} = \dots = p_{ii}^n \end{aligned}$$

For the expectation of W_i we have by definition (we put formally $F_i(0) = 0$):

$$\begin{aligned} m_i &= E(W_i) = \sum_{n=1}^{\infty} n[(1 - F_i(n-1)) - (1 - F_i(n))] \\ &= 1 + \sum_{n=1}^{\infty} (1 - F_i(n)) = \sum_{n=0}^{\infty} p_{ii}^n = \frac{1}{1 - p_{ii}} \end{aligned}$$

So,

$$p_{ii} = 1 - \frac{1}{m_i}, \quad i = 1, 2, \dots, K \quad (3.8)$$

To calculate the remaining p_{ij} , $j \neq i$, one can use assumption (a2). Let there be k_i states $\{r, s, \dots, z\}$ in the scheme attainable from state i in one step and let expert knowledge believe their likelihood to be in proportions $b_{ir}:b_{is}:\dots:b_{iz}$. We normalise these proportions in a way that $b_{ir} + b_{is} + \dots + b_{iz} = 1$ and call them *weights*. If we put now $b_{ij} = 0$ for any state j non-attainable from i in one step, then the transition probabilities

$$p_{ij} = b_{ij}(1 - p_{ii}), \quad j \neq i = 1, 2, \dots, K \quad (3.9)$$

obey both the stochastic conditions (Eqs. (3.1) and (3.2)) and the expert-given weight proportions.

If, in a particular case, all the transitions from state i to other k_i attainable states are equiprobable, then, for each i , all non-zero b_{ij} are equal, so that

$$p_{ij} = \frac{1}{m_i k_i}, \quad j \neq i \quad (3.10)$$

Note that, given a scheme of transitions, any reasonable choice of the time step insures $m_i \geq 1$ for any i ($m_i = 1$ implies $W_i \equiv 1$). If we presume $m_i > 1$, it follows that $p_{ii} > 0$ for any i . Mathematically, this means *regularity* of the Markov chain. The assumption that $m_i > 1$ is not crucial and can be relaxed (for transient states) without affecting the major results (see Section 5). Technically, one could increase m_i for any state i by reducing the time step of the chain, so that the assumption is not too restrictive.

The basic result for regular chains is the following (Karlin, 1968): there exists the limiting matrix $P(\infty) = \lim_{n \rightarrow \infty} P^n$ such that for any initial distribution $\mathbf{x}(0)$ we have

$$\mathbf{x}(n) = \mathbf{x}(0)P(n) \rightarrow \mathbf{x}(0)P(\infty) = \mathbf{x}^* \quad \text{as } n \rightarrow \infty \quad (3.11)$$

Vector $\mathbf{x}^*$ is called the *stationary distribution* due to the property $\mathbf{x}^*P(\infty) = \mathbf{x}^*$. It can be found by a standard routine for finding the eigenvector of the eigenvalue $\lambda = 1$ for matrix $P(1)$ (see, e.g. Roberts, 1976). If $P(1)$ has only one unitary eigenvalue, then $\mathbf{x}^*$ (Eq. (3.11)) can be easily shown to be actually independent of $\mathbf{x}(0)$ as all the rows of matrix $P(\infty)$ are identical in this case and equal to $\mathbf{x}^*$. Yet there may well be cases (rarely

addressed in applied manuals though) where $P(1)$ has more than one eigenvalues $\lambda_j = 1$, the limiting matrix $P(\infty)$ still having some regular pattern interpretable in terms of the succession scheme behind it (see, e.g. Denisenko et al., (1996) as an example of two climax states in the succession scheme for the forest-steppe zone and respectively two unitary eigenvalues in the spectrum of $P(1)$). A well-known practical approach to calculation of matrix $P(\infty)$ consists in multiplying matrix $P(1)$ by itself unless a regular pattern becomes visible in the product.

4. Continuous model formalism

When time is considered to be continuous, the stochastic and Markov properties are similar to those in the discrete formalism:

$$p_{ij}(t) \geq 0, \quad i, j = 1, \dots, K \quad (4.1)$$

$$\sum_{j=1}^K p_{ij}(t) = 1, \quad i = 1, \dots, K \quad (4.2)$$

$$P(s+t) = P(s)P(t), \quad t, s \geq 0 \quad (4.3)$$

Now t, s denote the continuous-time variables; the initial condition for Eq. (4.3) takes the form

$$\lim_{t \rightarrow 0+} P(t) = I \quad (4.4)$$

As well as in the discrete case, the state distribution $x(t)$ can be expressed in terms of the initial distribution $x(0)$ and matrix $P(t)$:

$$x(t) = x(0)P(t) \quad (4.5)$$

The matrix function $P(t)$ is continuous in t (Karlin, 1968). Furthermore, the following limits always exist:

$$\lim_{t \rightarrow 0+} \frac{1 - p_{ii}(t)}{t} = \lambda_i; \quad 0 \leq \lambda_i < \infty;$$

$$\lim_{t \rightarrow 0+} \frac{p_{ij}(t)}{t} = \lambda_{ij}; \quad i \neq j; \quad 0 \leq \lambda_{ij} < \infty,$$

where the λ values are interpreted as the transition probability densities. Theoretically, given a matrix $P(t)$, one can always construct the *transition density matrix*,

$$A = \begin{bmatrix} \lambda_{11} & \lambda_{12} & \dots & \lambda_{1K} \\ \vdots & \vdots & & \vdots \\ \lambda_{K1} & \lambda_{K2} & \dots & \lambda_{KK} \end{bmatrix}, \quad \lambda_{ii} = -\lambda_i$$

with the following property guaranteed by Eq. (4.2) for any finite Markov chain:

$$\lambda_i = \sum_{j=1, j \neq i}^K \lambda_{ij} \quad (4.6)$$

Due to this construction, the Kolmogorov–Chapman differential equations for transition probabilities take now the form

$$\begin{cases} P'(t) = P(t)A = AP(t), \\ P(0) = I, \end{cases}$$

with the solution $P(t) = \exp\{At\}$. Therefore, to calculate the transition probability matrix $P(t)$ for any $t > 0$, it is sufficient to know the matrix A . As well as in the discrete case, we use the expectations m_i to calculate A , relying upon the following mathematical ground.

Let W_i be the random variable defined in Section 3 and $F_i(t)$ be its probability distribution law. Similar to the ideas behind derivation of Eq. (3.8), we come from the Markov property to the following functional equation:

$$1 - F_i(t+s) = [1 - F_i(s)][1 - F_i(t)] \quad \forall s, t \geq 0,$$

w.r.t. unknown function $[1 - F_i(t)]$. The solution is well known to be exponential, whereby it follows that

$$\Pr\{W_i \leq t | X(0) = i\} = F_i(t) = 1 - \exp\{-c_i t\}$$

where c_i is a positive constant. What is also known for (and being advantageous of) continuous Markov chains is that $c_i = \lambda_{ii}$ (Chung, 1967), hence we have, as the expectation of the exponential distribution, $E[W_i] = 1/\lambda_i$, and

$$\lambda_i = 1/m_i \quad (4.8)$$

Altogether, we have the following construction. Given m_i , the average waiting times in the process states $i = 1, \dots, K$, the densities λ_i are immediately calculated by Eq. (4.8). From (Chung, 1967) it is known that the probability of transition to state j upon exit from state i ($i \neq j$) is equal to λ_{ij}/λ_i .

Since these probabilities have the same meaning as the weights b_{ij} for the discrete chain, it is natural to set $\lambda_{ij}/\lambda_i = b_{ij}$. Hence,

$$\lambda_{ij} = b_{ij} \cdot \lambda_i \quad (4.9)$$

Using Eq. (4.6), for equiprobable transitions, i.e. when $b_{ij} = 1/k_i$, we obtain thereafter

$$\lambda_{ij} = \lambda_i/k_i \quad (4.10)$$

for each j linked from the i and, setting the remaining entries to zero, the whole matrix A , which completely determines matrix $P(t)$ for any $t \geq 0$.

It is known (Ventzel and Ovcharov, 1991) that for any continuous-time chain the limiting matrix $P(\infty) = \lim_{t \rightarrow \infty} P(t)$ exists, such that for any initial distribution $\mathbf{x}(0)$ we have

$$\mathbf{x}(t) = \mathbf{x}(0)P(t) \rightarrow \mathbf{x}(0)P(\infty) = \mathbf{x}^* \text{ as } t \rightarrow \infty.$$

5. Comparison between the two formalisms

In view of the apparent similarity between two previous Sections, it is natural to consider now whether the discrete and continuous models yield identical results. In this Section we show the asymptotic equivalence between these two formalisms.

Let $P_d(n)$ and $P_c(t)$ be the transition probability matrices for the discrete and continuous cases respectively, constructed on a single set of m_i values. If m_i are expressed in time steps, then Eqs. (3.8) and (3.9) and Eqs. (4.8) and (4.9) result apparently in the following relationship among the parameters of the discrete chain and the continuous one:

$$P_d(1) = A + I. \quad (5.1)$$

Suppose that $m_i > 1$, i.e. the chain is regular, hence the major theorems on the limit distribution are applicable in the discrete and continuous cases. The following result holds true:

$$\lim_{n \rightarrow \infty} [P_d(n) - P_c(n)] = 0. \quad (5.2)$$

A simple proof is given in the Appendix. As a consequence, given an initial distribution $\mathbf{x}(0)$,

both models will produce the same stationary distribution $\mathbf{x}^*$.

There is one more property that the discrete and continuous descriptions have in common. To see it, we need some further notions. Let T_i denote the total time (number of steps) the discrete chain sojourns at state i . The state i is called *transient* if $E[T_i] < \infty$ and *ergodic* if $E[T_i] = \infty$. It is known from general theory (Kemeny and Snell, 1960) that:

(1) Any finite-state Markov chain has at least one ergodic state;

(2) Renumbering, if needed, the chain states, it is always possible to represent the transition probability matrix in the following form:

$$P_d(1) \begin{bmatrix} Q & R \\ 0 & A \end{bmatrix} \quad (5.3)$$

where submatrix Q (of size $q \times q$) contains only transitions between all pairs of transient states, while submatrix A (of size $(K-q) \times (K-q)$) corresponds to the ergodic states.

There always exists the *fundamental matrix*

$$N = [n_{ij}] = \sum_{k=0}^{\infty} Q^k = (I - Q)^{-1} \quad (5.4)$$

with the entries, $n_{ij} = E_i[T_j]$, representing the conditional expectation of T_j on condition that the chain comes out from state i .

In simpler words, the chain eventually leaves all the transient states and comes to the ergodic ones. The set of ergodic states is often called the *absorbing set*, because after reaching this set the chain can never leave it. For this reason, the ergodic states are normally interpreted as the climax ones. It is interesting to know how long it takes, on the average, to reach the climax states from any initial transient state i . Mathematically, we should find the expectation of the time $T_{\text{transient}}$ the chain sojourns totally at the relevant transient states starting at state i . With matrix N , the answer can be written immediately:

$$E_i(T_{\text{transient}}) = \sum_{j=1}^q n_{ij} \quad (5.5)$$

where q denotes the number of transient states.

In the Appendix we prove that Eq. (5.5) can be expanded to the continuous case too, in which matrix P is calculated formally by Eq. (5.1) and represented in the block form (Eq. (5.3)).

6. Example

Fig. 1 represents a scheme of succession through forest types in a mixed (spruce-broad-

$$P_{d,\min}(1) =$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & .7500 & .0625 & .0625 & .0625 & .0625 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & .9167 & 0 & 0 & 0 & .0833 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & .9167 & 0 & 0 & 0 & .0833 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & .9643 & 0 & 0 & 0 & .0357 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & .9750 & 0 & 0 & 0 & .0250 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & .9583 & 0 & 0 & 0 & .0417 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9750 & 0 & 0 & 0 & 0 & .0125 & .0125 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9750 & 0 & 0 & .0250 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9750 & 0 & 0 & 0 & 0 & .0250 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9583 & 0 & 0 & 0 & 0 & .0417 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9583 & 0 & 0 & 0 & .0417 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9583 & 0 & 0 & .0417 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .9583 & .0417 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

leafed) forest in Central Russia as a result of natural overgrowing on an abandoned tillage area (Korotkov, V.N., personal communication). As the first effort in the Markov-chain modelling technique, we consider an ideal course of succession with no disturbances; the general direction of changes is therefore unique: from the pioneer stage of succession to the climax one, which is considered to be a polydominant spruce-broad-leafed forest (Popadyuk et al., 1994). The chain has apparently 14 transient states and one absorbing state that corresponds to climax. Expert knowledge of how long a stage of succession can last (numbers of years in boxes of the scheme) until it turns into another one has rather the form of ranges than exact estimates of the mathematical expectations m_i . It makes sense in this case to model the fastest and the slowest courses of succession. The corresponding chains can be constructed by calculations as above for all the minimal, $D_{\min}$, and all the maximal, $D_{\max}$, values of m_i . If the time step is chosen to be $\Delta t = 5$ year (it cannot be greater than the shortest waiting

time among all stages), then the scheme of Fig. 1 suggests the following vectors of m_i -values:

$$D_{\min}$$

$$= [1 \ 4 \ 12 \ 12 \ 28 \ 40 \ 24 \ 40 \ 40 \ 40 \ 24 \ 24 \ 24 \ 24],$$

$$D_{\max}$$

$$= [1 \ 6 \ 12 \ 12 \ 28 \ 40 \ 24 \ 50 \ 50 \ 50 \ 24 \ 24 \ 24 \ 24].$$

In the discrete case, the matrix of one-step transition probabilities constructed by Eqs. (3.8) and (3.10) for vector $D_{\min}$ takes on the form:

The continuous chain is defined by its matrix $A_{\min}$ that refers to the matrix $P_{d,\min}(1)$ as shown in Eq. (5.1). Solving the initial-value problem for Eq. (4.7) yields thereafter the transition probability matrix

$$P_{c,\min}(t) = \exp\{A_{\min}t\},$$

that can be quantified further by means of the well-known formula

$$P(t) = \exp\{At\} = I + At + A^2t^2/2 + \dots$$

To illustrate the convergence ensuing from Eq. (5.2) for the two types of models, we calculate $x(n)$ and $x(t)$ by Eq. (3.7) and Eq. (4.5) respectively, at some discrete time instants $t = n = 0, 1, 2, \dots$, for the initial distribution $x(0) = [1, 0, \dots, 0]$, that corresponds to the whole area being originally at the pioneer stage of succession. In practice, one could monitor all kinds of distinguishable forest types, yet we confine, in the illustration, to the subclimax and climax stages (components 11–15). Fig. 2 shows dynamics of these components in both models plotted

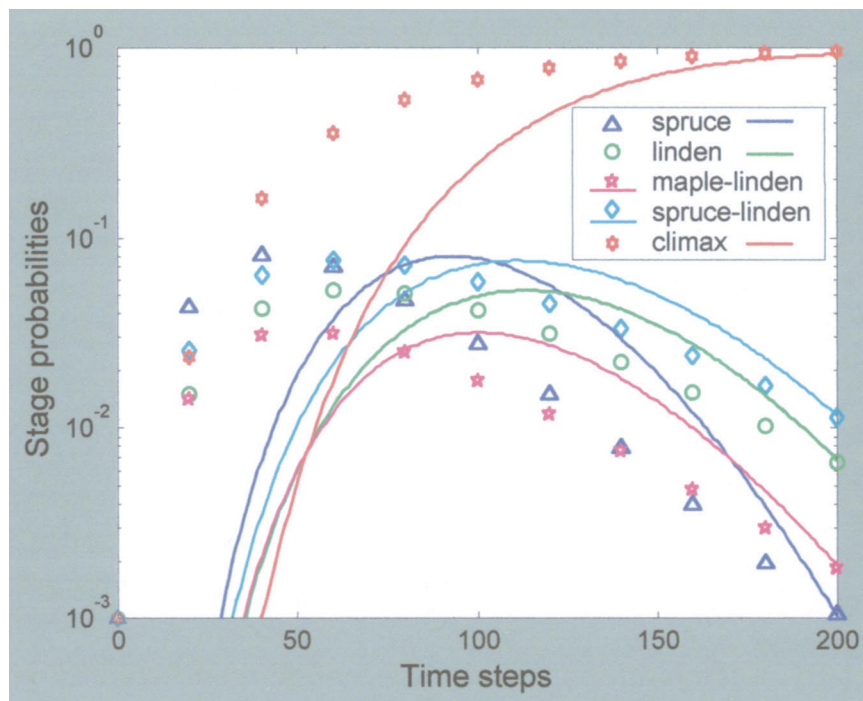


Fig. 2. Continuous and discrete model dynamics of the climax and subclimax components for the initial distribution $\mathbf{x}(0) = [1, 0, \dots, 0]$.

versus a single time axis in the log-linear scale, chosen to emphasise deviations which are hardly visible in the linear scale. Each subclimax curve is clearly seen to decline after a temporal increase, while the spruce trajectories walk away from the others, which looks natural due to the shorter waiting time of the spruce stage. Both common sense and mathematical results persuade that everything gets eventually into the climax stage in the ideal course of succession, i.e. $\mathbf{x}(\infty) = \mathbf{x}^* = [0, \dots, 0, 1]$, and both the discrete and continuous descriptions of that course do converge.

Calculated in the same way for the $D_{\max}$ set of m_i -values, the matrix $P_{d,\max}(1)$ has logically the same pattern as $P_{d,\min}(1)$, yet different numerical entries (not given) in those rows which correspond to differing components in vectors $D_{\max}$ and $D_{\min}$. The transition probability matrices $P_{d,\min}(1)$ and $P_{d,\max}(1)$ (or the transition density matrices $A_{\min}$ and $A_{\max}$ for the continuous chain)

thus define the potential range of variations in the rate of succession process.

For both discrete and continuous cases, Eqs. (5.3) and (5.4) produce the fundamental matrix N , whose elements n_{ij} estimate the number of time steps the succession stays at stage j if started from stage i . If, in particular, we use $P_{d,\min}(1)$ as P , then

$$N_{\min} = \begin{bmatrix} 1 & 4 & 3 & 3 & 7 & 10 & 6 & 10 & 10 & 10 & 6 & 6 & 3 & 9 \\ 0 & 4 & 3 & 3 & 7 & 10 & 6 & 10 & 10 & 10 & 6 & 6 & 3 & 9 \\ 0 & 0 & 12 & 0 & 0 & 0 & 24 & 0 & 0 & 0 & 24 & 0 & 0 & 0 \\ 0 & 0 & 0 & 12 & 0 & 0 & 0 & 40 & 0 & 0 & 0 & 0 & 12 & 12 \\ 0 & 0 & 0 & 0 & 28 & 0 & 0 & 0 & 40 & 0 & 0 & 24 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 40 & 0 & 0 & 0 & 40 & 0 & 0 & 0 & 24 \\ 0 & 0 & 0 & 0 & 0 & 0 & 24 & 0 & 0 & 0 & 24 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 40 & 0 & 0 & 0 & 0 & 12 & 12 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 40 & 0 & 0 & 24 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 40 & 0 & 0 & 0 & 24 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 24 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 24 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 24 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 24 \end{bmatrix}$$

Its row sums (multiplied by the time step length) now give the estimation (in years) of how long the succession stays in the transient stages if started from any earlier stage i :

$T_{\text{transient}}$

$= [440 \ 435 \ 300 \ 380 \ 460 \ 520 \ 240 \ 320 \ 320 \ 320 \ 1 \ 20 \ 120 \ 120 \ 120]^T$.

When calculated for matrix N_{max} (not cited), the estimation yields:

$T_{\text{transient}}$

$= [487.5 \ 482.5 \ 300 \ 430 \ 510 \ 570 \ 240 \ 370 \ 370 \ 37 \ 0 \ 120 \ 120 \ 120 \ 120]^T$.

So, it would take from 440 to 487 years, on the average, for the ideal course of succession to reach the climax stage from the pioneer one and respectively lesser times from the later stages.

Although some of these predictions might be regarded trivial by common sense relying on Fig. 1, like e.g. those for the pathways $3 \rightarrow 7 \rightarrow 11 \rightarrow 15$, 300 years, or $6 \rightarrow \dots \rightarrow 15$, 520–570 years, the predictions are not so obvious for the ramifying pathways to climax, especially if the alternative transitions were not be equiprobable.

7. Discussion

As can be seen in Section 3, the convergence behaviour of homogeneous chains, to say nothing of the ‘equilibrium’ terminology, fits surprisingly well to the Clementsian paradigm of succession theory suggesting a persistent motion to the climax state of vegetation which is in equilibrium with the environment. This conformity was noticed since the pioneers of Markov-chain modelling of successions (Horn, 1975; Usher, 1979) and has certainly been appealing to later followers. The opponents however could not accept a methodology “in which the future development of a system is determined by the present state and is independent of the way in which that state has developed” (Jeffers, 1978; page 89) — a widespread and vulgar (oversimplified) interpretation of the Markov property. Actually, it follows from Eq. (3.5) and the meaning of p_{ij} values that “the present state” does not determine completely “the future development of a system”: it rather does only the next step of the model and only in the sense that the *probability* distribution of the next state is independent of the current state prehis-

tory. In general, if an expert suggests a particular scheme of successional transitions in the form of a directed graph (like that in Fig. 1), then he/she already acts as if the Markov postulate has been accepted: for the stages where the graph does not ramify, the transition must be unique for any kind of models; at a ramifying stage, the total set of possible transitions is already defined by the very graph and it only remains to apply actual data/knowledge to estimate the probability distribution over that set. Therefore, the validity of the Markov postulate is rather a matter of the succession knowledge than of the way it is formalised in chain modelling.

Current theories of forest succession might rather suggest another motive to criticise the Markov-chain approach: while an absorbing chain predicts inevitable absorption in the absorbing state(s) to be associated with the climax one(s), the *mosaic concept* of age-space structure and dynamics of forest ecosystems (Borman and Likens, 1979; McIntosh, 1985; Smirnova et al., 1990; Korotkov, 1991; Popadyuk et al., 1994) treats the equilibrium state as a permanent cycling of spatial forest elements at different stages of succession (e.g. stage 15 in Fig. 1) sustained due to the balance between disturbance and regeneration processes (see also Logofet, 1999). This motive however could hardly discredit Markov chains since they are also able to model the above kind of asymptotic behaviour. For this purpose, the scheme of transitions should now include the disturbance links to the earlier stages that would provide for the directed graph (or its substantial part) to be strongly connected (Logofet, 1993). The task to reproduce an actual ratio among various stages in the limiting vector would thereafter pose a problem to estimate both the space ratio in the field and the temporal rates of disturbance — this problem might be solved, in the future projects, by means of the modern GIS-technologies developed in the time dimension (Acevedo et al., 1995). Markov models have long been known to be able to produce the non-absorbing asymptotic dynamics too (see, e.g. Jeffers, 1978).

Another, more profound, motive to criticise Markov models is the postulate of homogeneity,

i.e. the assumption that p_{ij} values are time-independent (Section 2). Prediction by a homogeneous model can be correct only to the extent that the invariance hypothesis holds true. The hypothesis is fairly dubious when the time horizon is long enough, yet a strict threshold can hardly be fixed here. It rather depends on the objectives of the modelling study. For example, if the study be aimed at effects of climate change, the invariance hypothesis would apparently not be true and the course of succession should depend on the changes in the key factors of the environment which the succession is known to depend upon. Those factors, like e.g. ground-water level and salinity for the flood-plain vegetation (Knyazkov et al., 1989, 1992) or the thermal and moistening factors in the forest-steppe zone (Logofet et al. 1997), while affecting the rate of successional processes, may well undergo some long-term trends induced by climate change. The resulting models (ibid.) represent *non-homogeneous* Markov chains: the transition matrix is constructed as a function, $P[E(t)]$, of the key factors $E(t)$, which summarises the knowledge and data about the succession under study. Loosing the algebraic elegance of their homogeneous prototypes, the non-homogeneous models now gain sensitivity to climate-sensitive parameters of the environment, hence acquire a specific prognostic ability.

Although the history of non-homogeneous Markov-chain models is not so long as homogeneous ones, it includes some case studies (ibid.; Logofet, 1999) and attempts to develop the methodological principles (Logofet and Denisenko, 1999). A crucial point in the methodology is finding the range of potential variations in matrix $P[E(t)]$, like those given by matrices $P_{\min}$, $P_{\max}$ and $A_{\min}$, $A_{\max}$ in Section 6 (see also Denisenko et al., 1996, and Logofet et al., 1997, for an example of how the range can be found from observations by a non-traditional usage of traditional models). The task to construct a model for a given data set, seeming to be quite a 'forward' task for the modeller, in mathematical terms, means solving a *backward problem* (finding a system of equations whose solution(s) reproduce the data). Backward problems are known to be typically *ill-posed* in the mathematical sense, i.e.

to have more than one solutions or/and diverging solutions for close original data. As shown in Sections 3 and 4, this is fortunately not the case for the Markov-chain formalisms: given a unique set of the average waiting times m_i and likelihood proportions b_{ij} for the ramifying transitions, the transition probability matrix is restored in a unique way, the backward problem thus being rather well-posed here.

This gives a mathematical ground to the proposed method of p_{ij} estimation, which differs drastically from what is used traditionally and based on substituting the frequency of an event within a set of observations for its probability within a model. Statistically, the latter approach implies registering all possible transitions in a great number of trials, which means quite a long history of observations, much longer than the life span of (even several generations of) the observers. Historical data can hardly meet the formal requirements which a confident statistical inference imposes on the number of 'repetitions of the trial': for instance, in a case study of forest successions in the State Nature Reserve 'Bryanskii Les' (Central Russia), with a seeming abundance of data in hand, only one half of the edaphotopes under study turned out to be supplied with a proper number of 'repetitions' (Pomaz et al., 1998).

A common trick to ride out the obstacle is to apply air/space imaging and to accumulate data for the statistics through the space dimension instead of the temporal one. In geobotanic terms, it means to substitute the *ecological series* (i.e. a sequence of ecosystems changing along an environmental gradient, see, e.g. Walter, 1979) for the successional (chronological) series. Extensively used also in demography and life sciences, this substitution is generalised in the mathematical notion of ergodicity. In random process theory, an *ergodic* process is such that a statistical characteristic of any realisation of the given process in time (of any trajectory) coincides with the characteristic over the whole ensemble of realisations (the state space) for any fixed moment of time. So, using observations in the space dimension to estimate the transition probabilities in time means accepting the *ergodic hypothesis*. This hypothesis may be true, like in the famous 'Indiana dunes'

where Cowles made his pioneer studies of changes in vegetation (Cowles, 1899; Odum, 1983) and where was a gradient apparently coincident with the historical process: while retreating to its present boundaries, Lake Michigan exposed successively newer and newer areas for sand dunes, so that increasingly older stages of successions could be found as one proceeded away from the shore. In other cases, matching the temporal gradient with an environmental one may not be so obvious, thus leaving the issue of ergodicity as a matter of modelling hypotheses, either explicit or implicit ones. Note that existence of ergodic sets of states in a stochastic process does not yet guarantee the process to be ergodic in the above sense. (More details on the ergodic hypothesis and ‘non-ergodic’ models can be found in Logofet, 1999).

The method we propose for p_{ij} estimation is apparently free of both the ergodic hypothesis in, and statistical requirements to, the data behind the model. The empirical evidence is rather supposed to be generalised in the expert views of the succession scheme and the rates at which the transitions occur. Mathematical analysis is thus separated from statistical problems inherent in estimation of m_i and b_{ij} values.

8. Conclusions

Traditional, time-homogeneous Markov chains serve an adequate didactic means to illustrate, either by their discrete formalism or continuous one, the classic paradigm of succession theory when expressed as a quantified scheme of successional transitions. As a straightforward formalisation, free of many technical assumptions that other dynamic models normally do, the homogeneous model gives numerical certainty to what the scheme alone could predict in rather qualitative terms. For example, the chain can predict the theoretical course of succession, the ratio between its final stages, and the mean time it would take to reach the final stage(s) under the hypothesis of invariance. It thus gives geobotanic knowledge the quantitative form capable of prediction. The homogeneous Markov chain can hardly become a

tool for realistic ecological prediction but can certainly be further developed to non-homogeneous, or even non-Markov, versions formalising the further knowledge of the succession under study. These kinds of models could be carefully calibrated in case studies using modern GIS-technologies to represent the mosaic of forest vegetation types and may be challenged to provide for the time dimension in those representations; they have certain potential to respond to this challenge.

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Appendix A

Statement 1.

$$\lim_{n \rightarrow \infty} [P_d(n) - P_c(n)] = 0. \quad (\text{A.1})$$

Proof. It is known (Karlin, 1968) that the stochastic matrix P_d for a regular Markov chain has one or more eigenvalues equal to one, and the corresponding eigenvectors are linearly independent. All the remaining eigenvalues lie inside the region $|z| < 1$. It follows that P_d can be represented, in the Jordan basis, as a matrix $J = CP_dC^{-1} =$

$$\begin{bmatrix} I_r & 0 \\ 0 & L \end{bmatrix}, \text{ where } I_r \text{ is the identity matrix } r \times r$$

($r \geq 1$) and matrix L contains only eigenvalues λ such that $|\lambda| < 1$. Therefore, $CP_d(n)C^{-1} = J^n \rightarrow$

$$J(\infty) = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ as } n \rightarrow \infty. \text{ For the continuous}$$

chain with matrix A calculated by Eq. (5.1) we have, in the same basis, $CAC^{-1} = J - I$, hence

$$\begin{aligned}
 CP_c(t)C^{-1} &= \exp\{(J - I)t\} \\
 &= \exp \begin{bmatrix} 0 & 0 \\ 0 & (L - I_{K-r})t \end{bmatrix} \xrightarrow{t \rightarrow \infty} J(\infty)
 \end{aligned}$$

which completes the proof.

Note that the proof gives also a method for calculating $P_d(\infty) = P_c(\infty) = C^{-1}J(\infty)C$.

Statement 2. In the continuous-time Markov chain

$$E_i(T_{\text{transient}}) = \sum_{j=1}^q n_{ij} \quad (\text{A.2})$$

where $[n_{ij}] = N = (I - Q)^{-1}$, and $q \times q$ -submatrix Q is defined by Eq. (5.1) and the block form Eq. (5.3).

Proof. It follows from Eq. (5.1) and Eq. (5.3) that matrix A takes on the following block form:

$$A = \begin{bmatrix} Q - I & R \\ 0 & A - I \end{bmatrix}.$$

Confined to the transient states, the Kolmogorov–Chapman equations $P'(t) = AP(t)$ can now be rewritten as follows:

$$\tilde{P}(t) = (Q - I)^{-1} \tilde{P}'(t) = -N \tilde{P}'(t). \quad (\text{A.3})$$

Consider a random variable $u(t)$ that is equal to 1 if $X(t) = j \leq q$ (any transient state) and to 0 otherwise. Obviously,

$$T_{\text{transient}} = \int_0^\infty u(t) dt.$$

The expectation of $T_{\text{transient}}$ is thereafter, in view of Eq. (A.3),

$$\begin{aligned}
 E_i[T_{\text{transient}}] &= E_i \left[\int_0^\infty u(t) dt \right] = \int_0^\infty E_i[u(t)] dt \\
 &= \int_0^\infty \sum_{j=1}^q p_{ij}(t) dt = \sum_{j=1}^q \int_0^\infty p_{ij}(t) dt \\
 &= \sum_{j=1}^q \left[\int_0^\infty \left(- \sum_{k=1}^q n_{ik} p'_{kj}(t) \right) dt \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \sum_{j=1}^q \left(- \sum_{k=1}^q n_{ik} p_{kj}(t) \right) \Big|_0^\infty \\
 &= \sum_{j=1}^q \left(\sum_{k=1}^q n_{ik} p_{kj}(0) \right) = \sum_{j=1}^q n_{ij}
 \end{aligned}$$

which completes the proof.

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